Template Practices Document

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1 Worked Solution Examples

Problems:

1.1. Example Uniform Investment Plan (C1 C2)

1.2. Example Linear Density Model (C2)

1.3. Example Table and Probability Summary (C3)

1.1 Example Uniform Investment Plan (C1 C2)

A variable V follows a uniform model on $[8, 16]$. Use the graph and density model to answer the tasks.

Questions/Tasks

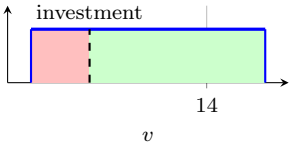
C1 Write the density function and support.
C2 Compute $P(10 \le V \le 14)$ and justify with area.

C1 - Write the density function and support

$$f_V(v) = \begin{cases} \dfrac{1}{16-8} = \dfrac{1}{8}, & 8 \le v \le 16, \\ 0, & \text{otherwise.} \end{cases}$$

$$8 \le v \le 16 \quad \longrightarrow \quad f_V(v) = \frac{1}{8} \quad \longrightarrow \quad \int_8^{16} f_V(v)\,dv = 1$$

C2 - Compute $P(10 \le V \le 14)$ and justify with area.



$$P(10 \le V \le 14) = \int_{10}^{14} \frac{1}{8}\, dv = \frac{1}{8}(14-10) = \frac{4}{8} = 0.5$$

The shaded area represents half of the support interval.

1.2 Example Linear Density Model (C2)

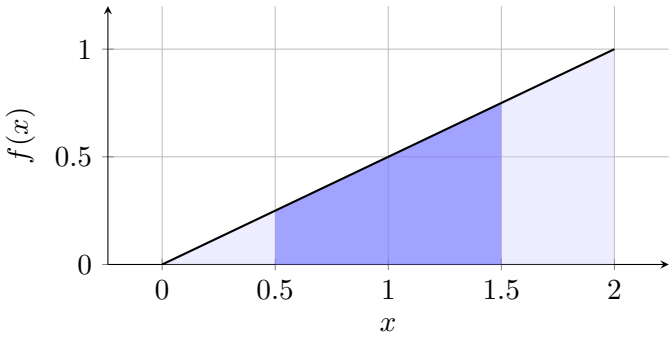
Suppose X has density $f(x) = kx$ for $0 \le x \le 2$.

Questions/Tasks

C2.1 Find the value of k.
C2.2 Compute $P(0.5 \le X \le 1.5)$.

C2.1 - Find the value of k.

$$1 = \int_0^2 kx\, dx = k\left[\frac{x^2}{2}\right]_0^2 = 2k \quad \Rightarrow \quad k = \frac{1}{2}$$



C2.2 - Compute $P(0.5 \le X \le 1.5)$.

$$P(0.5 \le X \le 1.5) = \int_{0.5}^{1.5} \frac{1}{2} x\, dx = \left[\frac{x^2}{4}\right]_{0.5}^{1.5} = \frac{1.5^2 - 0.5^2}{4} = 0.5$$

1.3 Example Table and Probability Summary (C3)

A summary table is shown for two intervals of a piecewise density.

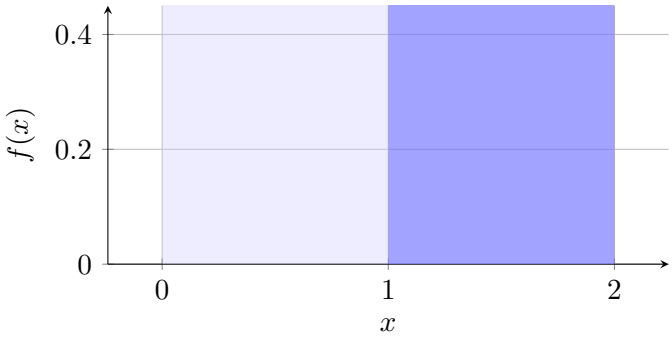
Questions/Tasks

C3.1 Complete the probability summary table.

C3.2 State one consistency check.

C3.1 - Complete the probability summary table.

Interval	Expression	Result
$[0, 1]$	$\int_0^1 0.4\,dx$	0.4
$[1, 2]$	$\int_1^2 0.6\,dx$	0.6
Total	$0.4 + 0.6$	1



C3.2 - State one consistency check.

A valid check is that all interval probabilities are nonnegative and add to 1.

2 Graph Command Examples

This section provides one representative command from each graph file in `book/preamble/graphs`.

Problems:

2.1. Example from `uniform_linear_probability_distributions.tex`

2.2. Example from `probability_distribution_graphs.tex`

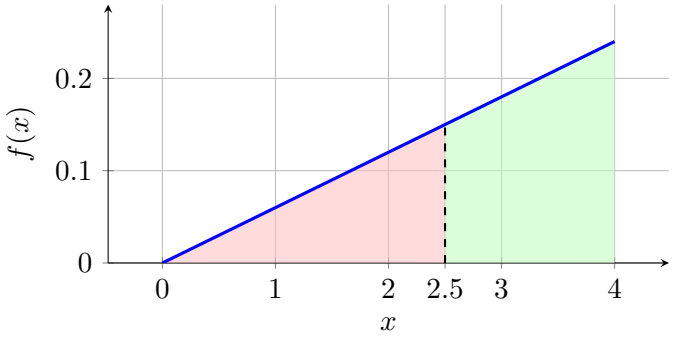
2.3. Example from `normal_distribution_graphs.tex`

2.1 Example from `uniform_linear_probability_distributions.tex`.

Questions/Tasks

C2 Plot the linear density and shade the region below the threshold.

C2 - Plot the linear density and shade the region below the threshold.

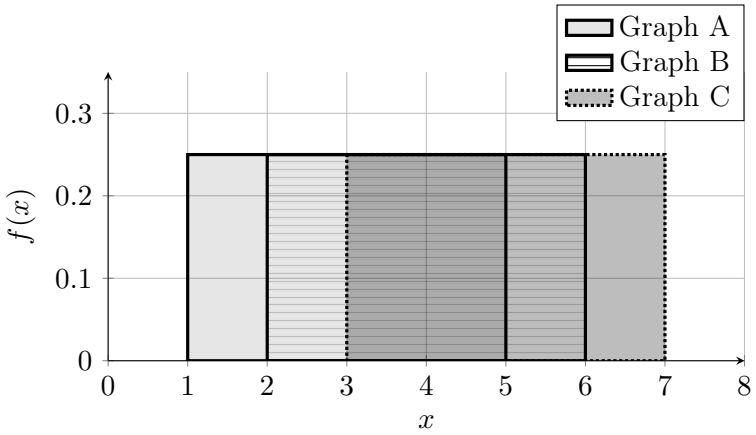


2.2 Example from `probability_distribution_graphs.tex`.

Questions/Tasks

C2 Plot the uniform density with three shaded intervals.

C2 - Plot the uniform density with three shaded intervals.



2.3 Example from `normal_distribution_graphs.tex`.

Questions/Tasks

C2 Plot the standard normal density and shade $P(Z \leq -0.75)$.

C2 - Plot the standard normal density and shade $P(Z \leq -0.75)$.

